Dhananjay Davhale

Python AI / LLM Engineer (RAG, FastAPI)

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Immediate joiner

Summary

Python AI Engineer with 4+ years of software experience and hands-on work building production RAG systems: document extraction, chunking, embeddings, hybrid retrieval (full-text + vector search with Qdrant) and LLM-based answer generation, served through async FastAPI services. Built Cortex, an organizational-memory AI platform (React + FastAPI), and Jessy, a production RAG assistant with chat-based statement upload for a wealth-management product. Strong backend foundation in PostgreSQL, Redis, Docker and AWS.

Technical Skills

AI / LLM: RAG pipelines, LLM application development, LangChain, Embeddings, Chunking strategies, Hybrid search, Semantic retrieval, Prompt design

Vector and Search: Qdrant, ChromaDB, PostgreSQL full-text search, JSONB filtering

Backend: Python, FastAPI, Pydantic, Async SQLAlchemy, Alembic, REST APIs, OAuth2, JWT

Data and Document Processing: PyMuPDF, python-docx, Pandas, NumPy, ETL pipelines

Databases: PostgreSQL, MySQL, Redis

Cloud and DevOps: Docker, Docker Compose, AWS (Lambda, API Gateway, S3, IAM, Cognito, CloudWatch, Secrets Manager), Git

Projects

Cortex - Organizational Memory AI Platform | Personal Project (React, FastAPI, Async SQLAlchemy, PostgreSQL, Qdrant, Redis, Docker)

Teams ship thousands of features, but the requirements, validations and owners behind each one stay buried in emails and documents. Cortex lets anyone ask in plain English and get answers from those sources.

- Built the full product solo: React frontend (login, document upload, chat interface) and FastAPI backend, so users can ask who requested a feature and what its requirements and validations were, instead of digging through emails.
- Built document extraction pipelines with PyMuPDF, python-docx and Pandas, and implemented chunking for semantic search.

- Implemented hybrid retrieval (PostgreSQL full-text search, JSONB filtering, Qdrant vector search) so exact matches like names and IDs and meaning-based questions both return correct results.
- Managed schema evolution with Alembic migrations and containerised the stack with Docker Compose.

Jessy - AI Assistant for a Wealth-Management Product | Asset Vantage, in production (Python, FastAPI, RAG, LLM, Qdrant, PostgreSQL, Redis)

- Built a RAG chatbot that answers user questions on mutual funds, bonds, bond ROC, how to transact, reports and new features, cutting repetitive support queries.
- Kept answers current with feature mappings and snapshot-based explanations, reducing user confusion and support load.
- Conceived and built chat-based statement upload during a hackathon (Top 3): users drop a PDF into the chat and the Statement Processing Engine reads it and posts the transactions into the right investment vehicle, replacing long manual entry forms.
- Designed, built and shipped solo to production as a premium-subscriber feature.

Work Experience

Software Engineer, Asset Vantage (product company, wealth management) | Pune | Feb 2023 - Jul 2026

- Built the RAG chatbot and AI knowledge features (LLMs, embeddings, Qdrant vector search, document processing) on a live production product.
- Built the Statement Processing Engine, a large module that converts financial statements into structured data.
- Delivered 50+ backend features individually, including Document Preview and Bond Master, using Python, FastAPI and MySQL/PostgreSQL.
- Built serverless services on AWS Lambda and API Gateway with Cognito, IAM, Secrets Manager and CloudWatch.
- Improved report-generation workflows, improving processing efficiency by ~20%.
- Awarded the Bright Beginner Award (2023).

Fullstack Developer, Exousia Web | Jun 2021 - Jan 2022

- Built web applications and backend functionality with PHP, JavaScript and MySQL, including API integrations.

Education

MCA (Computers), Dr. D. Y. Patil Institute of Engineering, Management and Research, Pune | 2021 - 2023

B.C.A. (Computer Applications), Indrayani Vidya Mandir's, Pune | 2018 - 2021

Certifications

Introduction to Generative AI | Amazon Web Services | Google Advanced Data Analytics